TALHA’S PHYSICS ACADEMY

XI PHYSICS IMPORTANT M.C.Qs FOR 2024

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1. [MLT-2] is dimensional formula of:

(a) strain (b) force (c) displacement (d) pressure

2. Which of the following pair has the same dimension?

(a) moment of inertia and torque (b) surface tension and force

(c) impulse and momentum (d) specific heat and latent heat

3. The unit of solid angle is

(a) Second (b) Steradian (c) Kilogram (d) Candela

4. Dimensions of kinetic energy is the same as that of:

(a) Acceleration (b) Velocity (c) Work (d) Force

5. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10-3 are:

(a) 5, 1, 2 (b) 5, 1, 5 (c) 5, 5, 2 (d) 4, 4, 2

6. Which among the following is the supplementary unit:

(a) Mass (b) Time (c) Solid angle (d) Luminosity

7. The quantity having the same unit in all system of unit is:

(a) mass (b) time (c) length (d) temperature

8. The velocity of a particle at an instant is 10 m/s and after 5sec the velocity of particle is 20m/s. The velocity 3 sec before in m/s is:

(a) 8 (b) 4 (c) 6 (d) 7

9. At the top of a trajectory of a projectile, the acceleration is:

(a) maximum (b) minimum (c) zero (d) g

10. At what angle the range of projectile becomes equal to the height of projectile?

(a) 65° (b) 45° (c) 76° (d) 30°

11. The angle at which dot product becomes equal to the cross product is:

(a) 65° (b) 45° (c) 76° (d) 30°

12. To get a resultant displacement of 10m, two displacement vectors of magnitude 6m and 8m should be combined:

(a) Parallel (b) Antiparallel (c) At an angle of 45° (d) Perpendicular to each other

13. A ball is thrown upwards with a velocity of 100 m/s. It will reach the ground after:

(a) 10 sec (b) 20 sec (c) 5 sec (d) 40 sec

14. Two projectiles are fired from the same point with the same speed at angles of projection 60° and 30° respectively. Which one of the following is true?

(a) The range will be same (b) Their maximum height will be same

(c) Their landing velocity will be same (d) their time of flight will be same

15. The rate of change of linear momentum of a body is called a:

(a) Linear force (b) Angular force (c) Power (d) Impulse

16. The term mass refers to the same physical concept as:

(a) weight (b) inertia (c) force (d) acceleration

17. The motion of rocket in the space is according to the law of conservation of:

(a) Energy (b) linear momentum (c) mass (d) angular momentum

18. A bomb of mass 12 kg initially at rest explodes into two pieces of masses 4kg and 8kg. The speed of 8kg mass is 6 m/s. The kinetic energy of 4 kg mass is:

(a) 32 J (b) 48 J (c) 114 J (d) 288 J

19.The kinetic energy of body of mass 2kg and momentum of 2 Ns is:

(a) 1 J (b) 2 J (c)3 J (d) 4 J

20. For the same kinetic energy, the momentum is maximum for:

(a) an electron (b) a proton (c) a deuteron (d) alpha particle

21. If momentum is increased by 20% then K.E increases by:

(a) 44% (b) 55% (c) 66% (d) 77%

22.The angular momentum vector of Earth about its rotation axis, due to its daily rotation is, directed

(a) tangent to the equator towards east (b) tangent to the equator towards the west

(c) north (d) towards the Sun.

23. A stone of 2 kg is tied to a 0.50 m long string and swung around a circle at angular velocity of 12 rad/s. The net torque on the stone about the center of the circle is

(a) 0 N.m (b) 6 N.m (c) 12 N.m (d) 72 N.m

24. A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms

(a) his angular velocity increases. (b) his angular velocity remains same

(c) his rotational inertia decreases (d) his angular momentum remains the same.

25. A space station revolves around the earth as a satellite, 100 km above the Earth's surface. What is the net force on an astronaut at rest inside the space station?

(a) equal to her weight on earth (b) a little less than her weight on earth

(c) less than half her weight on earth (d) zero (she is weightless)

26. If the external torque acting on a body is zero, then its

(a) angular momentum is zero (b) angular momentum is conserved

(c) angular acceleration is maximum (d) rotational motion is maximum

27. One radian is about

(a) 25° (b) 37° (c) 45° (d) 57°

28. The rotational inertia of a wheel about its axle does not depend upon its

(a) diameter (b) mass (c) distribution of mass (d) speed of rotation

29.A body of mass 5 kg is moving with a momentum of 10 kg m/s. A force of 0.2 N acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy is:

(a) 2.8 J (b) 3.2 J (c) 3.8 J (d) 4.4 J

30.The kinetic energy of a light and a heavy object is same. Which object has maximum momentum?

(a) Light object (b) Heavy object (c) Both have same momentum (d) N.O.T

31.A body fallen from height h. After it has fallen a height h/2, its will possess:

(a) only potential energy (b) only kinetic energy

(c) half potential half kinetic energy (d) more kinetic and less potential

32. Which of the following quantity can be calculated by multiplying force and velocity?

(a) acceleration (b) power (c) torque (d) work

33. The minimum velocity given to an object so that it emerges out from the gravitational field of earth is about

(a) 11.2 km/s (b) 15.3 km/s (c) 5 km/s (d) 9.8 km/s

34. The absolute potential energy of an object depends on:

(a) The object's mass and height (b) The object's mass and speed

(c) The object's shape and size (d) The object's color and temperature

35. The escape velocity of a planet depends on which of the following factors?

(a) The mass of the planet only (b) The radius of the planet only

(c) Both the mass and the radius of the planet (d) The density of the planet

36. A completely submerged object always displaces its own

(a) weight of fluid (b) volume of a fluid (c) density of a fluid (d) Area of fluid

37. The pressure exerted on the ground by a man is greatest when:

(a) he stands with both feet flat on (b) he stands flat on one foot

(c) he stands on the toes of one foot the ground (d) he lies down on the ground

38. In a stationary homogeneous liquid:

(a) pressure is the same at all points (b) pressure depends on the direction

(c) pressure is independent of any atmospheric pressure on the upper surface of the liquid

(d) pressure is the same at all points at the same level

39. One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston:

(a) increases by 2ΔP (b) increases by Δp/2 (c) increases by Δp (d) increases by 4Δp

40. In a vacuum, an object has

(a) No buoyant force (b) no mass (c) no weight (d) none of these

41. The pressure at the bottom of a pond does NOT depend on

(a) Water density pond (b) the depth of the pond (c) the surface area (d) None of these

42. Salt water has greater density than freshwater. A boat floats in both freshwater and saltwater. The buoyant force on the boat in salt water is that in freshwater.

(a) equal to (b) smaller than (c) larger than (d) same as

43. For an incompressible fluid, the flow rate is

(a) equal for all surfaces. (b) constant throughout the pipe.

(c) greater for the larger parts of the pipe. (d) none of the above

44. As the speed of a moving fluid increases, the pressure in the fluid

(a) increases (b) remains constant

(c) decreases (d) may increase or decrease, depending on the viscosity

45. If the cross-sectional area of a pipe decreases, what happens to the fluid velocity?

(a) Increases (b) Decreases (c) Remains the same (d) Depends on the fluid density

46. A sky diver falls through the air at terminal velocity. The force of air resistance on him is

(a) half his weight (b) equal to his weight

(c) twice his weight (d) Cannot be determined from the information given.

47. Wind speeding up as it blows over the top of a hill

(a) Increases atmospheric pressure there. (b) decreases atmospheric pressure there.

(c) doesn’t affect atmospheric pressure there. (d) equal’s atmospheric pressure.

48. A fluid is undergoing “incompressible” flow. This means that:

(a) the pressure at a given point cannot change with time (b) the velocity at a given point cannot change with time

(c) the velocity must be the same everywhere (d) the density cannot change with time or location

49.The equation of continuity for fluid flow can be derived from the conservation of:

(a) energy (b) mass (c) volume (d) pressure

50. The minimum charge on an object cannot be less than:

(a)1.6×10^{-19} C (b) 3.2×10^{-19}C (c) 9.1×10^{9}C (d) No definite value exist

51. Two charges are placed at a certain distance. If the magnitude of each charge is doubled the force will become

(a)1/4th of its original value (b) 4 times of its original value

(c) 1/8th of its original value (d) 8 times of its original value

52. Which of the following can be deflected while moving in the electric field?

(a)neutron (b) photon (c) electron (d) (a) and (b)

53. The flux through a flat surface of are "A" in a uniform electric field "E" is maximum when the surface area is:

(a) Parallel to E (b) perpendicular to E (c) placed 45° to E (d) placed 60° to E

54. The force between two charges placed in air is F. if air is replaced by a medium of relative permittivity €, then force is reduced to:

(a)$F\epsilon_r$ (b) F/ϵ_r (c) ϵ_r/F (d) $\epsilon\epsilon_r$

55. The negative gradient of the potential is:

(a) potential energy (b) voltage (c) electric field intensity (d) electric flux

56. The electric flux through a plane area will be half of its maximum value when area is held at angle of ________ with electric field

(a) 30° (b) 60° (c) 45° (d) 90°

57. The capacitance of a capacitor is NOT influenced by

(a) Plate thickness (b) Plate area (c) Plate separation (d) Nature of the dielectric

58. What is the value of capacitance of a capacitor which has a voltage of 4V and has 16C of charge?

(a) 2F (b) 4F (c) 6F (d) 8F

59. Energy stored in the capacitor is:

(a) E=1/2 CV (b) $E=\frac{1}{2} CV^2$ (c) $E = CV^2$ (d) E = ½ CV

60. The time constant of a series RC circuit consisting of 100uF capacitor in series with a 100 resistor is.

(a) 0.1 s (b) 0.1 ms (c) 0.01 s (d) 0.01 ms

61. The charging of a capacitor through a resistance follows

(a) linear law (b) square law (c) exponential law (d) none of the above

62. When the total charge in a capacitor is doubled, the energy stored

(a) remains the same (b) is doubled (c) is halved (d) is quadrupled

63. Capacitor blocks:

(a) alternating current (b) direct current

(c) both alternating and direct current (d) neither alternating nor direct current

64. The resistance of a superconductor is:

(a) finite (b) infinite (c) changes with every conductor (d) zero

65. Reciprocal of resistance is called:

(a) conductance (b) resistivity (c) resonance (d) capacitance

66. The graphical representation of Ohms law is:

(a) parabola (b) hyperbola (c) ellipse (d) straight line

67. Internal resistance is the resistance offered by:

(a) Capacitor (b) resistor (c) Conductor (d) source of emf

68. What is a potentiometer primarily used for?

(a) Measuring electric current (b) Measuring electric charge

(c) Measuring potential difference (voltage) (d) Measuring electric resistance

69. A heat-sensitive device whose resistivity changes with the change in temperature is called:

(a) conductor (b) resistor (c) thermistor (d) thermometer

70. A wire of uniform area of cross-section A length L and resistance R is cut into two parts. The resistivity of each part:

(a) Becomes zero (b) is halved (c) Is doubled (d) remains same

71. Two simple pendulums A and B with same lengths, and equal amplitude of vibrations, but the mass of A is twice the mass of B, their period are TA and TS and energies are EA and ES respectively. Choose the correct statement.

(a) TA = TS and EA< EB (b) TA < TB and EA > EB (c) TA < TB and EA < EB (d) TA = TB and EA < EB

72. In order to double the period of a simple pendulum:

(a) Its length should be doubled (b) Its length should be quadrupled

(c) The mass should be doubled (d) The mass should be quadrupled

73. A simple harmonic oscillator has amplitude A and time period t. Its maximum speed is:

(a) $\frac{4A}{t}$ (b) $\frac{2A}{t}$ (c) $\frac{4\pi A}{t}$ (d) $\frac{2\pi A}{t}$

74. A spring attached by a load of weight W is vibrating with a period T. If the spring is divided in four equal parts and the same load is suspended from one of these parts, the new time period is:

(a) T/4 (b) 2T (c)T/2 (d) 4T

75. A child swinging on a swing in sitting position, stands up, then the time period of the swing will:

(a) Increase (b) decrease

(c) remains the same (d) increases if the child is long and decreases if the child is short

76. If a body oscillates at the angular frequency ω of the driving force, then the oscillations are called:

(a) Forced oscillations (b) Coupled oscillations (c) Free oscillations (d) Maintained oscillations

77. A simple harmonic oscillator with a natural frequency ω_N is force to oscillate with a driving frequency ω d. The Resonance occurred when:

(a) ωN < ωd (b) ωN > ωd (c) ωN = ωd (d) ωN ≅ ωd

78. A heavily damped system has a fairly flat resonance curve in:

(a) An acceleration time graph (b) An amplitude frequency graph

(c) Velocity time graph (d) Distance-time graph

79. The speed v of a wave represented by y = A sin(wt -kx) is:

(a)k/ω (b) ω/k (c) ω k (d) l/ωk

80. Two sound waves are y = A sin(wt —kx) and y = A cos(wt —kx). The phase difference between the two waves is:

(a) $\pi/2$ (b) π /4 (c) π (d) 0

81. If va, vh and vm are the speeds of sound in air, hydrogen gas, and a metal at the same temperature, then:

(a) $v_a > v_h > v_m$ (b) $v_m > v_h > v_a$ (c) $v_h > v_m > v_a$ (d) $v_h > v_a > v_m$

82. The speed of sound in air at STP is 332 m/s. If the air pressure becomes double at the same temperatue, the speed of sound become:

(a) 1382 m/s (b) 664m/s (c) 332 m/s (d) 166 m/s

83. The length of a pipe closed at one end is L. In the standing wave whose frequency is 7 times the fundamental frequency, what is the closest distance between nodes?

(a)(1/14) L (b) (1/7) L (c) (2/7) L (d) (4/7) L

84. The speed of sound in a gas in which two waves of wavelength 50 cm and 50.4 cm produce 6 beats per second is:

(a) 338 m/s (b) 350 m/s (c) 378 m/s (d) 400 m/s

85. The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 minutes, then its wave length is:

(a) 13.85 m (b) 25.3 m (c) 41.5 m (d) 57.2 m

86. Huygens's conception of secondary waves

(a)helps us to find the focal length of a thick lens (b) is a geometrical method to find a wavefront

(c)is used to determine the velocity of light (d) is used to explain the polarization of light

87. Interference fringes are produced using monochromatic light of same intensity from a double slit screen. If the intensity of light emerging from one of the slit is reduced, the effect on interference pattern will be

(a)All the dark and bright fringes become brighter.

(b) All the dark and bright fringes become darker.

(c)Bright fringes become brighter and dark fringes become darker.

(d) Bright fringes become darker and dark fringes become brighter.

88. In Young's experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be;

(a) 4 times (b) 1/4 times (c) doubled (d) unchanged

89. A ray of light passes from air into water. Striking the surface of the water with an angle of incidence 45°. Which of these quantities change as the light enters the water.

i) Wavelength ii) frequency iii) speed of propagation iv) direction of propagation

(a) i and ii only. (b) iii and iv only.

(c) i, iii, and iv only. (d) all of them.

90. A hill separates a television (TV) transmitter from a house. The Transmitter cannot be seen from the house but still the TV in the house has good reception. What wave phenomena make it possible?

(a) Coherence of waves (b) Diffraction of waves (c) Interference of waves (d) Refraction of waves

91. Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed. Which effects is observed by replacing the grating with one that has more lines per millimeter?

(a)Number of maxima decreases with decrease in angle between first and second order maxima.

(b) Number of maxima decreases with increase in angle between first and second order max inn.

(c)Number of maxima increases with decrease in angle between first and second order maxima.

(d) Number of maxima increases with increase in angle between first and second order maxima.

92. Optically active substances are those substances which

(a)produce polarized light (b) rotate the plane of polarization of polarized light

(c)produce double refraction (d) convert a plane polarized light into circularly polarized light

93. Plane polarized light is passed through a Polaroid. On viewing through the Polaroid we find that when Polaroid is given one complete rotation about the direction of light

(a)The intensity of light gradually decreases to zero and remains at zero.

(b) The intensity of light gradually increases to maximum and remains at maximum.

(c)There is no change in the intensity of light.

(d) The intensity of light varies such that it is twice maximum and twice zero.

94. In Radio and Television broadcast, the information signal is in the form of:

(a) analog signal (b) digital signal

(c) Both analog & digital signals (d) neither analog nor digital signal

95. A communication channel consists of:

(a) transmission line only (b) optical fibre only

(c) free space only (d) All of them

96. Voltage signal generated by a microphone is:

(a) digital in nature (b)analog in nature

(c) hybrid in nature (d) consists of bits & bytes

97. As compared to sound waves frequency of radio waves is:

(a) higher (b) equal (c) lower (d) may be higher or lower

98. The process of superimposing signal frequency on the carrier wave is known as:

(a) transmission (b)detection (c) reception (d)modulation

99. Process of mapping the sampled analog voltage values to discrete voltage levels is called:

(a) sampling (b)sampling frequency (c) quantizing (d)encoding

100. AM is used for broadcasting because:

(a) it requires less transmitting power compare with other systems

(b) it is more noise immune than other modulation system

(c)No other modulation can provide the necessary bandwidth faithful transmission

(d) its use avoids receiver complexity